


Lancaster University College
at Beijing Jiaotong University

Digital Signal Processing

Zhou Hang

Discrete Fourier Transform

- Definition of the Discrete Fourier Transform
- Properties of the Discrete Fourier Transform
- Calculating Linear Convolution using DFT
- Analyzing Signal Spectra using DFT
- Calculating DFT using MATLAB

Utilize the DFT to compute linear convolution

- The steps for calculating linear convolution by DFT
- The linear convolution of a long sequence and a short sequence

Utilize the DFT to compute linear convolution



$$y[k] = x[k] * h[k]$$

The number of multiplications for linear convolution is $N \times M$, where $x[k]$ has a length of N and $h[k]$ has a length of M.

How to implement fast computation of linear convolution?

Utilize the DFT to compute linear convolution

By utilizing the circular convolution property of DFT, linear convolution can be calculated using DFT.

If the length of the sequence $x[k]$ is N and the length of the sequence $h[k]$ is M, then the circular convolution of $L > N+M-1$ points is equivalent to the linear convolution of $x[k]$ and $h[k]$.

$$x[k] * h[k] = x_1[k] \textcircled{L} h_1[k] \xleftrightarrow{\text{DFT}} X_1[m]H_1[m]$$

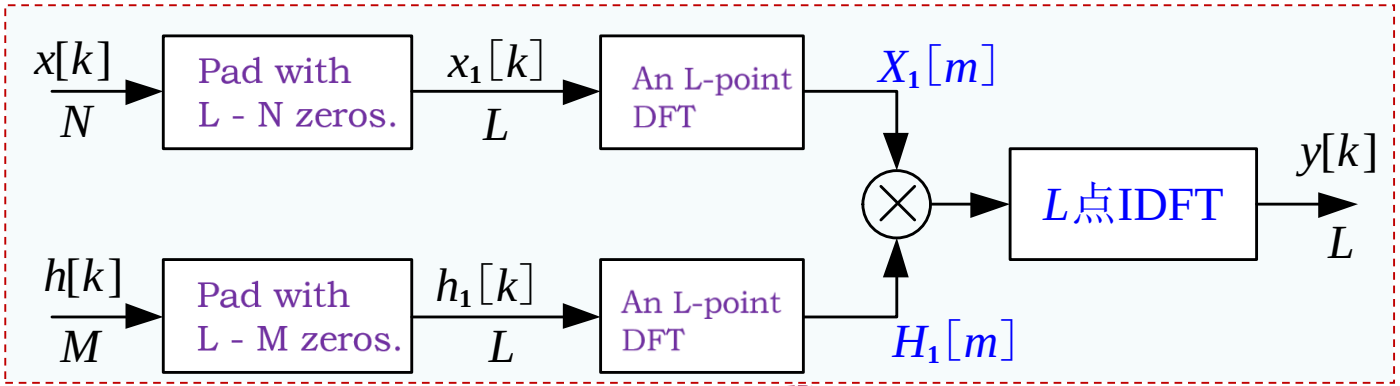
$$x[k] * h[k] = \text{IDFT}\{X_1[m]H_1[m]\}$$

Utilize the DFT to compute linear convolution

The length of the sequence $x[k]$ is N, and the length of the sequence $h[k]$ is M.

The length of the sequence $x_1[k]$ is L, and the length of the sequence $h_1[k]$ is also L.

$$x[k] * h[k] = \mathrm{IDFT}\{X_1[m]H_1[m]\}$$



The total number of multiplications is

$$3\times\frac{L}{2}\log_2 L + L$$

$L \geq N+M-1$

Example: Given $x[k]= \{1, 2, 3, 4\}$, $h[k]=\{5, 6, 7\}$, to calculate the linear convolution of $x[k]$ and $h[k]$ using DFT。

```
x = [1 2 3 4];     h = [5 6 7];

L = length(x)+length(h)-1; % Determine the length for zero padding

XE = fft(x, L);                % Compute the DFTs by zero-padding
HE = fft(h, L);

y = ifft(XE.*HE);              % Determine the IDFT of the product
```

```
y =    5    16    34    52    45    28
```

Utilize the DFT to compute linear convolution

Comparison of multiplication operation counts.

$$N = 2^{16}, M = 2^{16} \qquad L=N+M=2^{17}$$

Direct computation of linear convolution. $N \times M = 2^{32}$

Computation of linear convolution using DFT. $3 \times \frac{L}{2} \log_2 L + L = 53 \times 2^{16}$

The ratio of multiplication counts between the two methods.

$$\frac{2^{32}}{53 \times 2^{16}} = \frac{2^{16}}{53} \approx 2^{10}$$

Because there is a fast algorithm for DFT called FFT(Fast Fourier Transform), Using DFT (FFT) to compute linear convolution can significantly reduce the number of multiplications.

Utilize the DFT to compute linear convolution

Comparison of Multiplication Operation Counts

$N = 2^{16}, M = 2^4$ $L \geq N+M-1$， Take $L=2^{17}$

Perform Direct Linear Convolution $N \times M = 2^{20}$

Perform Linear Convolution via DFT $3 \times \frac{L}{2} \log_2 L + L = 53 \times 2^{16}$

The ratio of the number of multiplications between the two.

$$\frac{2^{20}}{53 \times 2^{16}} = \frac{2^4}{53} \approx 0.3$$

Using DFT to calculate the convolution of a long sequence with a short sequence actually increases the number of multiplications.

The linear convolution of a long sequence and a short sequence

Directly using DFT to calculate the convolution of a long sequence with a short sequence has the following drawbacks:

(1) The signal must be fully input before computation can proceed, resulting in excessive delay;

(2) Memory requirements are high.

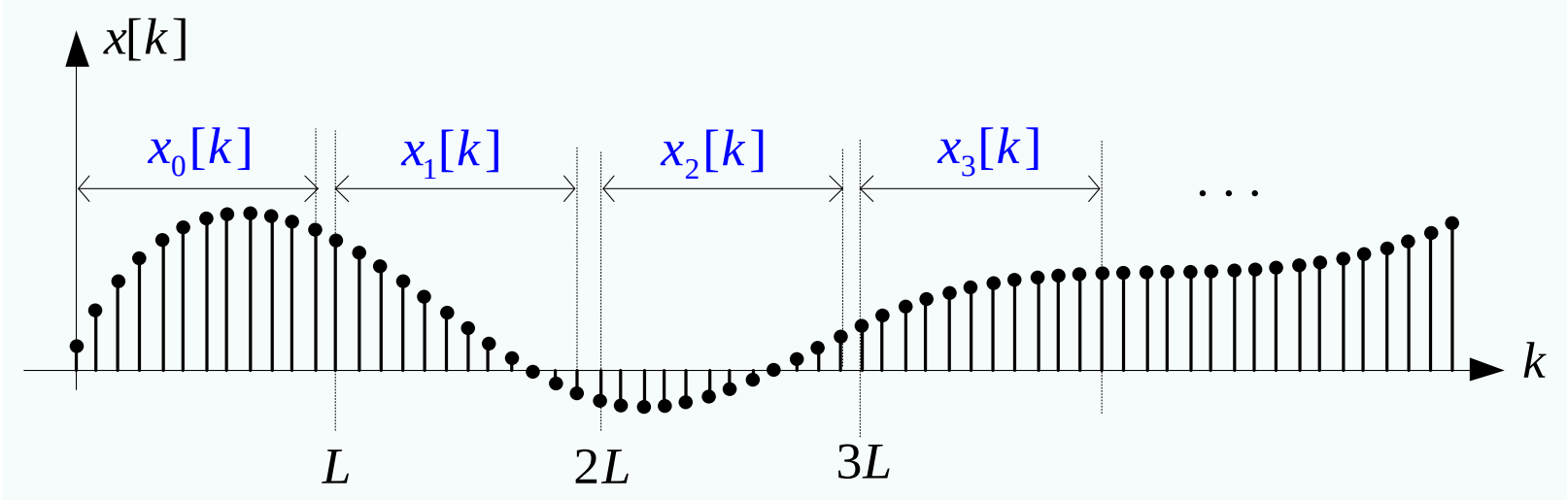
(3) The algorithm is not very efficient.

Use segmented convolution.

1. Overlap-Add Method
2. Overlap-Save Method

Overlap-Add Method

Divide the long sequence $x[k]$ into several short sequences of length L.



Definition $$x_n[k]=\begin{cases} x[k+nL] & 0\le k\le L-1 \\ 0 & \text{else} \end{cases}$$

$$x[k]=\sum_{n=0}^{\infty} x_n[k-nL]$$

Overlap-Add Method

$$y[k] = x[k] * h[k] = \sum_{n=0}^{\infty} x_n[k - nL] * h[k] = \sum_{n=0}^{\infty} y_n[k - nL]$$

$$y_n[k] = x_n[k] * h[k]$$

$$x[k] = \sum_{n=0}^{\infty} x_n[k - nL]$$

The length of $h[k]$ is M

$y_0[k]$ non-zero range. $0 \le k \le L + M - 2$

$y_1[k\text{-}L]$ non-zero range. $L \le k \le 2L + M - 2$

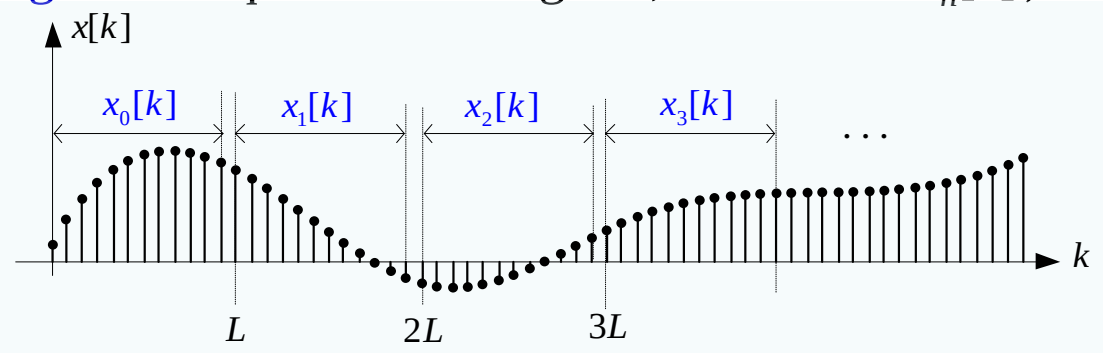
Sequence $y_0[k]$, $y_1[k\text{-}L]$ overlapping part. $L \le k \le L + M - 2$

The number of overlapping points. $L+M\text{-}2\text{-}L+1=M\text{-}1$

Add the $M-1$ overlapping points of two adjacent segments in sequence to obtain the final result of the linear convolution.

Overlap-Add Method

The Overlap-Add Method for calculating the linear convolution of a long sequence $x[k]$ and a short sequence $h[k]$:Divide the long sequence $x[k]$ into several non-overlapping short sequences of length L, denoted as $x_n[k]$;



(1) Calculate the linear convolution of $x_n[k]$ with $h[k]$;

(2) Add the M-1 overlapping points of adjacent segments to obtain the final result of the linear convolution.

Overlap-Add Method

Example: Given the sequence $x[k]=k+2, 0\le k\le 12$, $h[k]=\{1,2,1\}$ use the Overlap-Add Method to calculate the linear convolution, taking $L=5$.

Solution: $x[k]=\{2, 3, 4, 5, 6,\ 7, 8, 9, 10, 11,\ 12, 13, 14\}$ $L=5$

$x_0[k]=\{2, 3, 4, 5, 6\}$ $x_1[k]=\{7, 8, 9, 10, 11\}$ $x_2[k]=\{12, 13, 14, 0, 0\}$

$y_0[k]=\{2, 7, 12, 16, 20, 17, 6\}=x_0[k]*h[k]$

$y_1[k]=\{7, 22, 32, 36, 40, 32, 11\}=x_1[k]*h[k]$

$y_2[k]=\{12, 37, 52, 41, 14, 0, 0\}=x_2[k]*h[k]$

Overlap-Add Method

Example: Given the sequence $x[k]=k+2, 0\le k\le 12$, $h[k]=\{1,2,1\}$ use the Overlap-Add Method to calculate the linear convolution, taking $L=5$.

Solution:

$y_0[k]=\{2, 7, 12, 16, 20, 17, 6\}= x_0[k]*h[k]$

$y_1[k]= \{7, 22, 32, 36, 40, 32, 11\}=x_1[k]*h[k]$

$y_2[k]= \{12, 37, 52, 41, 14, 0, 0\}=x_2[k]*h[k]$

$y_0[k]=\{2, 7, 12, 16, 20, 17, 6\}$

$y_1[k-L]= \{7, 22, 32, 36, 40, 32, 11\}$

$y_2[k-2L]= \{12, 37, 52, 41, 14, 0, 0\}$

$\boldsymbol{M}=3$

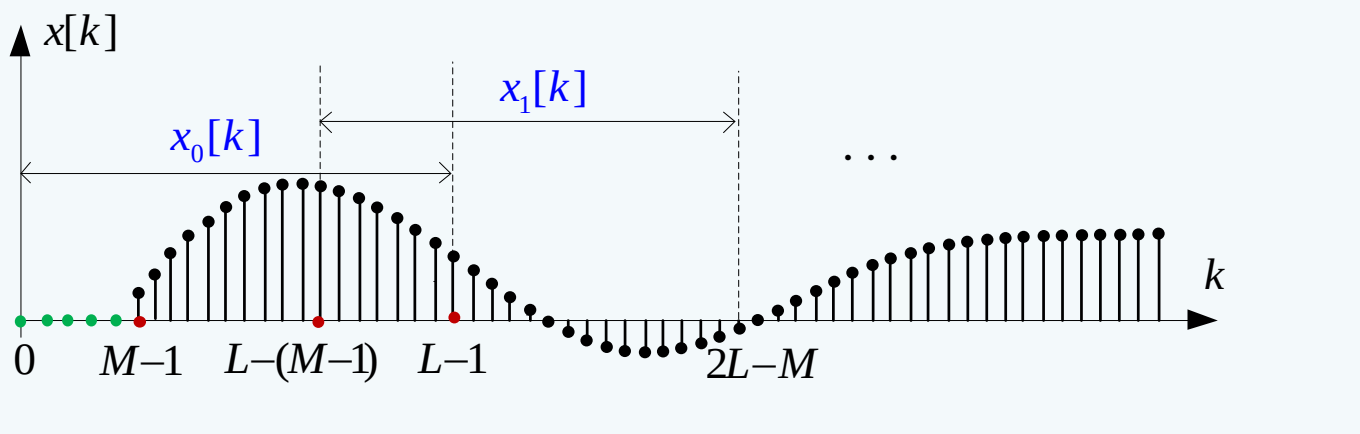
Adjacent segments have M – 1=2overlapping points that are added together.

$y[k]=\{2, 7, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 41, 14\}=x[k]*h[k]$

Overlap-Save Method

The Overlap-Save Method for calculating the linear convolution of a long sequence $x[k]$ and a short sequence $h[k]$:

(1) Divide the long sequence $x[k]$ into several short sequences of length L , denoted as $x_n[k]$, Each segment is overlapped by M-1 points with the previous one, and the first sequence is preceded by M-1 zeros;



Overlap-Save Method

The Overlap-Save Method for calculating the linear convolution of a long sequence $x[k]$ and a short sequence $h[k]$:

(2) Calculate the L-point circular convolution $y_n[k]$ of $x_n[k]$ with $h[k]$;

(3) Delete the parts of each $y_n[k]$ where $0 \le k < M-1$, then concatenate the remaining parts of adjacent segments in sequence to obtain the final result of the linear convolution.

$y_0[k]$=**{11, 4, 2, 7, 12}** $y_1[k]$=**{23, 17, 16, 20, 24}**

$y[k]$=**{2, 7, 12, 16, 20, 24}** M=3

Overlap-Save Method

Example: Given the sequence $x[k]=k+2, 0\le k\le 12$, $h[k]=\{1,2,1\}$ use the Overlap-Add Method to calculate the linear convolution, taking $L=5$.

Solution: $x[k]=\{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14\}$ $M=3$

$x_0[k]=\{0, 0, 2, 3, 4\}$ $x_1[k]=\{3, 4, 5, 6, 7\}$ $x_2[k]=\{6, 7, 8, 9, 10\}$

$x_3[k]=\{9, 10, 11, 12, 13\}$ $x_4[k]=\{12, 13, 14, 0, 0\}$

$y_0[k]=x_0[k]\textcircled{5}h[k]=\{\cancel{11, 4}, 2, 7, 12\}$ $y_1[k]=x_1[k]\textcircled{5}h[k]=\{\cancel{23, 17}, 16, 20, 24\}$

$y_2[k]=x_2[k]\textcircled{5}h[k]=\{\cancel{35, 29}, 28, 32, 36\}$ $y_3[k]=x_3[k]\textcircled{5}h[k]=\{\cancel{47, 41}, 40, 44, 48\}$

$y_4[k]=x_4[k]\textcircled{5}h[k]=\{\cancel{12, 37}, 52, 41, 14\}$ In $0\le k<M-1$,delete $y_n[k]$

$y[k]=x[k]*h[k]=\{2, 7, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 41, 14\}$

Thanks

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude.